COHOMOLOGY OF QUASI-COHERENT MODULES VANISHES ABOVE THE AFFINE STRATIFICATION NUMBER

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 21645-21693, immediately following the statement at 21638-21643.

Editorial note. Seven cross-chapter pointers (Cohomology of Schemes, Cohomology, Properties, Morphisms) plus the chapter's own definition are recorded in assembly.external_references.

Editorial note. The reduction to finite type modules supported on Z is displayed in the span before the support trick is applied.

Allowed standard prerequisites (editorial).

- Definition of the affine stratification number. chapter definition definition-affine-stratification-number, cited at proof line 21650. Supplies the inductive structure: an affine open U whose complement carries a stratification number one smaller.
- Vanishing of quasi-coherent cohomology on an affine scheme. Cohomology of Schemes, lemma coherent-lemma-quasi-coherent-affine-cohomology-zero, cited at proof lines 21648-21649. Provides the base case n = 0.
- Relative affine vanishing and affine cohomology comparison. Cohomology of Schemes, lemmas coherent-lemma-relative-affine-vanishing and coherent-lemma-relative-affine-cohomology, cited in the proof. Transfers the computation to the complement Z and identifies H^p(X, G) with H^p(Z', G).
- Colimits of quasi-coherent modules of finite type. Cohomology, lemma cohomology-lemma-quasi-separated-cohomology-colimit and Properties, lemma properties-lemma-quasi-coherent-colimit-finite-type, cited in the proof. Reduces the general quasi-coherent F to finite type modules supported on Z.

Source reference. Original reference 01XB: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 089W: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01XC: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 01FF: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 0F2V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21569–21569. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 056H: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PG: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01LA: focused-seeds/geometry/S9-morph-vanishing-affine-strat-number/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let X be a nonempty quasi-compact and quasi-separated scheme with affine stratification number n. Then $H^p(X, \mathcal{F}) = 0$, $p > n$ for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$.*

Proof. We will prove this by induction on the affine stratification number n. If $n = 0$, then X is affine and the result is Cohomology of Schemes, Lemma 01XB. Assume $n > 0$. By Definition 0F2V there is an affine scheme U and an affine open immersion $j : U \to X$ such that the complement Z has affine stratification number $n-1$. As U and j are affine we have $H^p(X, j_*(\mathcal{F}|_U)) = 0$ for $p > 0$, see Cohomology of Schemes, Lemmas 089W and 01XC. Denote $\mathcal{K}$ and $\mathcal{Q}$ the kernel and cokernel of the map $\mathcal{F} \to j_*(\mathcal{F}|_U)$. Thus we obtain an exact sequence

$$0 \to \mathcal{K} \to \mathcal{F} \to j_*(\mathcal{F}|_U) \to \mathcal{Q} \to 0$$

of quasi-coherent $\mathcal{O}_X$-modules (see Schemes, Section 01LA). A standard argument, breaking our exact sequence into short exact sequences and using the long exact cohomology sequence, shows it suffices to prove $H^p(X, \mathcal{K}) = 0$ and $H^p(X, \mathcal{Q}) = 0$ for $p \geq n$. Since $\mathcal{F} \to j_*(\mathcal{F}|_U)$ restricts to an isomorphism over U, we see that $\mathcal{K}$ and $\mathcal{Q}$ are supported on Z. By Properties, Lemma 01PG we can write these modules as the filtered colimits of their finite type quasi-coherent submodules. Using the fact that cohomology of sheaves on X commutes with filtered colimits, see Cohomology, Lemma 01FF, we conclude it suffices to show that if $\mathcal{G}$ is a finite type quasi-coherent module whose support is contained in Z, then $H^p(X, \mathcal{G}) = 0$ for $p \geq n$. Let $Z' \subset X$ be the scheme theoretic support of $\mathcal{G} \oplus \mathcal{O}_Z$; we may and do think of $\mathcal{G}$ as a quasi-coherent module on Z', see Morphisms, Section 056H. Then Z' and Z have the same underlying topological space and hence the same affine stratification number,

namely $n-1$. Hence $H^p(X,\mathcal{G}) = H^p(Z',\mathcal{G})$ (equality by Cohomology of Schemes, Lemma 089W) vanishes for $p \geq n$ by induction hypothesis. □